This technical report provides the detailed pseudocode of the proposed greedy algorithms and the proofs of the theoretical results. Algorithm 1 (Followers) describes the procedure for computing the follower set induced by an anchored edge. Algorithm 2 (GIA) presents the greedy incremental anchoring strategy, which iteratively selects the edge–increment action with the largest follower gain per unit budget. Algorithm 3 (GPA) accelerates GIA by using candidate reduction, follower-ratio upper bounds, and intermediate-result reuse, while preserving the same greedy selection criterion.

A Pseudo code of Followers

Algorithm 1: Followers(G, e, δ_e, s, C_s)

```
PQ ← queue(); // sort by coreness-layer ascending order
F ← ∅;
w(e) ← w(e) + δ_e;
for each x ∈ {u, v} if x ∉ C_s do
    PQ.add(x);
while PQ ≠ ∅ do
    x ← PQ.pop();
    w⁺(x) ← SupportWeightedDegree(G, x);
    if w⁺(x) ≥ s then
        F ← F ∪ {x};
        for each y ∈ N(x, G) and x ≺ y do
            if y ∉ C_s and y ∉ F then
                PQ.add(y);
    else
        Q ← queue();
        Q.push(x);
        while Q ≠ ∅ do
            y ← Q.pop();
            if y ∈ F then
                F ← F \ {y};
            for each z ∈ N(y, G) ∩ F do
                w⁺(z) ← w⁺(z) − w(y, z);
                if w⁺(z) < s then
                    Q.push(z);
w(e) ← w(e) − δ_e;
return F
```

Algorithm description. Algorithm 1 computes the follower set induced by an anchored edge via a localised simulation of the s-core peeling process. Starting from the anchored endpoints, the algorithm explores only nodes that are reachable through neighbours with higher coreness–layer pairs, following the upstair order established by the onion-layer structure. A node is retained as a follower if its support weighted degree meets the s-core threshold; otherwise, it is discarded. Discarding a node may reduce the support of neighbouring retained followers, potentially triggering further removals. This cascading update exactly mirrors the behaviour of the s-core decomposition, but is restricted to the structurally affected region. The process terminates when no further admissible nodes remain, and the surviving candidates constitute the final follower set.

B Pseudo code of GIA

Algorithm description. Algorithm 2 presents the baseline greedy incremental anchoring procedure of GIA. Starting from the initial s-core, the algorithm iteratively evaluates candidate edges whose endpoints are not both contained in the current s-core. For each candidate edge, GIA determines the required weight increment using the MEE rule, computes the induced follower set, and evaluates the corresponding follower ratio. At each iteration, the edge achieving the largest follower ratio is selected, its weight is increased accordingly, and the s-core is updated. This process repeats until the available budget is exhausted, yielding a greedy baseline for comparison.

Algorithm 2: GIA(G, b, s)

```
G' ← G, A ← ∅, sum ← 0;
C_s ← s-core(G');
while sum < b do
    E* ← C(V, 2) \ C(C_s, 2);
    e* ← none, δ* ← 0, max_FR ← 0;
    for each e ∈ E* do
        δ_e ← computeDelta(G', s, e);
        if sum + δ_e ≤ b then
            F(e, δ_e) ← Followers(G', e, δ_e, s, C_s);
            FR(e) ← |F(e, δ_e)|/δ_e;
            if max_FR < FR(e) then
                e* ← e, δ* ← δ_e;
                max_FR ← FR(e);
    w_G'(e*) ← w_G'(e*) + δ*;
    sum ← sum + δ*;
    A ← A ∪ (e*, δ*);
    C_s ← s-core(G');
return A
```

C Pseudo code of GPA

Algorithm description. Algorithm 3 presents the pseudocode of GPA, an accelerated variant of GIA that reduces the candidate search space while preserving the same greedy selection criterion. GPA exploits the locality of follower propagation by organising nodes outside the s-core into connected components and maintaining, for each component, the best edge–weight increment candidate pair identified so far. At each iteration, GPA selects the current best candidate across all components. It then explores candidate edges whose endpoints belong to different components, guided by node-level and edge-level upper bounds that allow early termination and safe pruning. Only candidate pairs that cannot be ruled out by these bounds are evaluated using the same follower computation as in GIA. After applying the selected edge reinforcement, GPA updates the component structure. If the reinforced edge connects two distinct components, they are merged and their best candidate is recomputed; otherwise, only the affected component is updated. In both cases, the s-core is recomputed locally within the affected components. This process repeats until the budget is exhausted, yielding the same greedy outcome as GIA with fewer candidate evaluations.

D Proof of Theorem 1

PROOF. We reduce the Maximum Coverage (MC) problem, which is known to be NP-hard unless P = NP [2], to the WASCO problem. The MC problem is defined as follows: given a collection of sets $\{T_1, T_2, \ldots, T_t\}$ containing elements from a universe $\{e_1, e_2, \ldots, e_d\}$

Algorithm 3: GPA(G, b, s)

```
G' ← G, A ← ∅, sum ← 0;
L ← list(), M ← map(); // maps each component to its best (e, δ_e, FR)
C_s ← s-core(G');
M ← buildComponents(G',b,s);
while sum < b do
    e*, δ*, max_FR ← arg max_{(e,δ_e,FR)∈M} FR;
    for u ∈ V \ C_s and sum + (s − c(u)) ≤ b do
        U(u) ← UpperBound(G', u, s);
        L.append(u);
    sort(L); // by upper bound decreasing order
    for u ∈ L do
        if max_FR > 2 · U(u) then
            break;
        for v ∈ L s.t. v appears after u do
            if u.component = v.component then
                continue;
            if max_FR > U(u) + U(v) then
                break;
            if max_FR > U(u, v) then
                continue;
            else
                e ← (u, v);
                δ_e ← computeDelta(G', s, e);
                if sum + δ_e ≤ b then
                    F(e, δ_e) ← Followers(G', e, δ_e, s, C_s);
                    FR(e) ← |F(e, δ_e)|/δ_e;
                    if max_FR < FR(e) then
                        e* ← e, δ* ← δ_e;
                        max_FR ← FR(e);
    w_{G'}(e*) ← w_{G'}(e*) + δ*;
    sum ← sum + δ*;
    A ← A ∪ (e*, δ*);
    updateComponents(G',b,s,sum,M);
return A
```

and a budget b', the objective is to select b' sets such that the number of unique elements covered is maximised.

To construct an instance of the WASCO problem, consider an arbitrary instance I_{MC} of the MC problem with sets $\{T_1, \ldots, T_t\}$ and elements $\{e_1, \ldots, e_d\} = \bigcup_{i=1}^{t} T_i$, as illustrated in the top of Figure 1. We define an instance $I_{\text{WASCO}} = (G, s, b)$ as follows:

- The graph G consists of three components: a part M representing the sets T_i, a part N representing the elements e_j, and a set of $(d+2)$-cliques.
- For each set T_i, create a corresponding set of nodes M_i in M, where $M_i = \{u_{i,1}, u_{i,2}, \ldots, u_{i,d+2}\}$. The nodes $\{u_{i,1}, \ldots, u_{i,d+1}\}$ form a $(d+1)$-clique, and $u_{i,d+2}$ is connected to $\{u_{i,1}, \ldots, u_{i,d}\}$.
- For each element e_j, create a corresponding node v_j in N. Add an edge between v_j and $u_{i,j}$ for every set T_i such that $e_j \in T_i$.

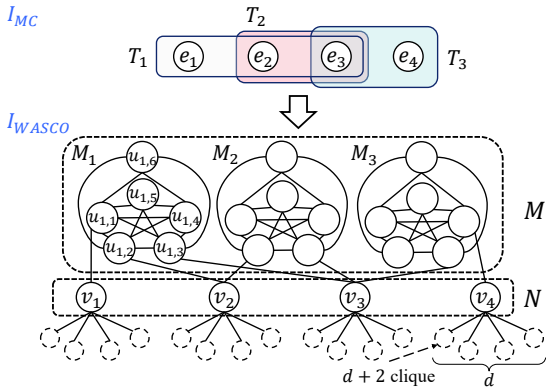


Figure 1: Proof of NP-hardness

- For each v_j, add d disjoint $(d+2)$-cliques, and connect v_j to one node in each clique.
- Set $s = d + 1$ and $b = b'$.
- Every edge in G has a weight of 1.

This construction can clearly be completed in polynomial time. Initially, during the s-core decomposition, all nodes in M and N are removed, leaving only the $(d+2)$-cliques. This happens because:

- Nodes $u_{i,d+1}$ and $u_{i,d+2}$ in M_i have degree d, which is less than $s = d + 1$, and are therefore removed.
- Removing $u_{i,d+1}$ and $u_{i,d+2}$ causes the remaining nodes $u_{i,j}$ in M_i to fail the s-core condition, as their degree drops below s.
- All v_j nodes in N are similarly removed because their connections to M_i are lost.

As a result, only the $(d+2)$-cliques remain in the s-core after this initial decomposition. To maximise the s-core size, an optimal solution is to add an edge with a weight of 1 between $u_{i,d+1}$ and $u_{i,d+2}$ in M_i. Adding this edge corresponds to selecting the set T_i in the MC problem. Note that an optimal solution could instead distribute the budget across edges between distinct sets in M (e.g., adding $(u_{i,d+2}, u_{j,d+1})$ and $(u_{j,d+2}, u_{i,d+1})$, possibly with fractional weights summing to the same budget). However, any such allocation yields no more followers than concentrating the same budget on a single intra-set edge $(u_{i,d+1}, u_{i,d+2})$, since a node enters the s-core only once its weighted degree reaches $s = d + 1$, and splitting the budget across two edges raises each node's weighted degree by less than committing the full weight to one. When the edge is added:

- All nodes in M_i are included into the s-core because the degree of $u_{i,d+1}$ and $u_{i,d+2}$ becomes $d + 1$.
- All nodes v_j connected to M_i are also included in the s-core, as their degree conditions are satisfied.

Since the optimal way to maximise the s-core is to add edges corresponding to specific T_i sets so as to include as many e_i elements as possible in the s-core, this process directly mirrors the objective of the MC problem to select sets that maximise the coverage of elements. Thus, selecting M_i that are more densely connected to uncovered v_j aligns with the optimisation goals of the MC problem.

Therefore, the MC problem is reducible to the WASCO problem, implying that $I_{MC} \leq_P I_{\text{WASCO}}$. As the MC problem is NP-hard, this proves the NP-hardness of the WASCO problem. □

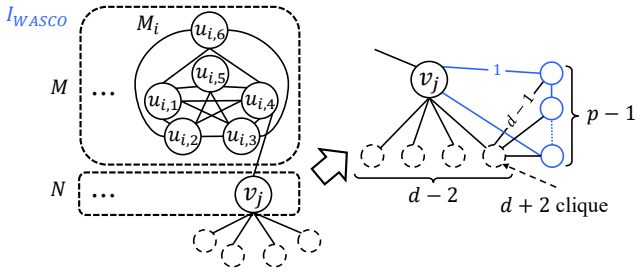


Figure 2: Gadget for the inapproximability reduction

E Proof of Theorem 2

Proof. We give a gap-preserving reduction from the Maximum Coverage (MC) problem. It is known that, for any $\varepsilon > 0$, MC cannot be approximated within a factor of $(1 - 1/e + \varepsilon)$ in polynomial time, unless P = NP [1].

We start from the construction in the proof of Theorem 1 and apply the same two modifications to each node in part N (Figure 2). For each element node v_j, we (i) reduce the number of disjoint $(d+2)$-cliques attached to v_j from d to $d-2$, and (ii) replace v_j by a cycle on p nodes by adding $p-1$ new nodes connected to v_j with edges of weight 1. Moreover, each of the $p-1$ new cycle nodes is connected to one node of a (randomly chosen) attached $(d+2)$-clique by an edge of weight $d-1$. We set $s = d+1$ as in Theorem 1. Let p be an integer chosen as a polynomially bounded function of the MC instance size; its value will be specified at the end of the proof.

Let G be the resulting WASCO instance, and let $f(A) = |C_s(G_A)|$ denote the size of the s-core after applying an anchoring solution A with total budget b. As in Theorem 1, without any anchoring, all nodes in M and N are removed by the s-core decomposition, and only the union of the $(d+2)$-cliques remains in the s-core. Denote by B the number of nodes in this always-retained clique part; B is fully determined by the MC instance and is independent of A.

Now consider any feasible anchoring solution A that uses exactly b budget units (if it uses fewer, we can add arbitrary weight increments on edges between already-retained clique nodes to exhaust the budget without changing the s-core size). By the same argument as in Theorem 1, anchoring an edge within a set gadget M_i corresponds to selecting the set T_i, which makes all $d+2$ nodes of M_i enter the s-core; since exactly b sets are selected, this contributes an additive term $(d+2)b$ to $f(A)$.

Let $\mathrm{cov}(A)$ be the number of elements covered by the corresponding b selected sets in the MC instance. By construction, $\mathrm{cov}(A)$ is exactly the number of element nodes v_j that enter the s-core under A. We next show that if a node v_j enters the s-core, then all p nodes on the cycle containing v_j also enter the s-core. Indeed, once v_j is retained, every other cycle node has weighted degree $2 + (d-1) = d+1 = s$, coming from its two cycle neighbours (weight 1 each) and its incident edge of weight $d-1$ to a clique node that is always retained. Hence, each cycle node satisfies the s-core condition in the induced subgraph consisting of the always-retained cliques together with this cycle. Therefore, the inclusion of v_j contributes an additional p nodes to the s-core.

Putting the above together, we obtain the following exact relation for every feasible A:

$$f(A) = B + (d+2)b + p \cdot \mathrm{cov}(A).$$

Let $\mathrm{OPT}_{\mathrm{MC}}$ be the optimal MC value, and let $\mathrm{OPT}_{\mathrm{W}}$ be the optimal WASCO objective. From the relation above,

$$\mathrm{OPT}_{\mathrm{W}} = B + (d+2)b + p \cdot \mathrm{OPT}_{\mathrm{MC}}.$$

Assume, for contradiction, that there exists a polynomial-time $(1 - 1/e + \varepsilon)$-approximation algorithm for WASCO. Let $\rho = 1 - 1/e + \varepsilon$, and let A be the solution returned by this algorithm. Then

$$B+(d+2)b+p\cdot\mathrm{cov}(A) = f(A) \geq \rho\,\mathrm{OPT}_{\mathrm{W}} = \rho\big(B+(d+2)b+p\cdot\mathrm{OPT}_{\mathrm{MC}}\big),$$

which implies

$$\mathrm{cov}(A) \geq \rho\,\mathrm{OPT}_{\mathrm{MC}} - \frac{(1-\rho)\big(B+(d+2)b\big)}{p}.$$

Choose

$$p \;\geq\; \frac{2\big(B+(d+2)b\big)}{\varepsilon},$$

which is polynomially bounded in the instance size. Since $\mathrm{OPT}_{\mathrm{MC}} \geq 1$ for any non-trivial MC instance, the additive term is at most $\varepsilon/2$, and thus

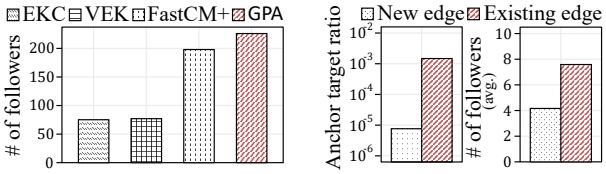
$$\mathrm{cov}(A) \;\geq\; (1 - 1/e + \varepsilon/2)\,\mathrm{OPT}_{\mathrm{MC}}.$$

This yields a polynomial-time $(1 - 1/e + \varepsilon/2)$-approximation for MC, contradicting the inapproximability of MC unless P = NP [2]. Therefore, WASCO cannot be approximated within a factor of $(1 - 1/e + \varepsilon)$ in polynomial time for any $\varepsilon > 0$, unless P = NP. □

F Proof of Theorem 3

Proof. Suppose that f is submodular. Then, for any two sets A and B, it must hold that $f(A)+f(B) \geq f(A\cup B)+f(A\cap B)$. Consider a graph consisting of four nodes $\{u, v, x, y\}$ and two edges (u, x) and (v, y), each with a weight of 1. Let $s = 2$, if $A = \{\langle (u, v), 1\rangle\}$ and $B = \{\langle (x, y), 1\rangle\}$, $f(A) + f(B) = 0 < f(A \cup B) + f(A \cap B) = 4$. □

G Additional Experiment



(a) $k = 20, b = 200$ **(b) Anchored edge analysis**

Figure 3: EQ1-4. Effectiveness on the Facebook

EQ1-4. Effectiveness (unweighted graph). We further evaluate GPA on the unweighted Facebook dataset [3] ($|V| = 4,039$, $|E| = 88,234$, $d_{\mathrm{avg}} = 43.7$). All compared methods, including EKC [5], VEK [6], and FastCM+ [4], are originally designed for unweighted graphs. To enable a fair comparison, we treat the unweighted graph as a special case of our weighted formulation by assigning unit weight to every edge, and follow the same parameter settings ($k = 20$, $b = 200$) as in FastCM+. As shown in Figure 3(a), GPA can achieve the largest number of followers among all compared methods under

the same budget. This result indicates that, even when restricted to the unweighted setting, GPA is capable of operating effectively in practice. Figure 3(b) further analyses the anchoring behaviour of GPA. After normalising by the size of the corresponding feasible action spaces, edge reinforcement is selected more frequently than new edge insertion. Moreover, reinforcement actions tend to yield a higher average number of followers per anchored edge. These observations suggest that strengthening existing connections can be a more effective anchoring strategy than edge insertion, even when all edges have uniform weight.

References

[1] Uriel Feige. 1998. A threshold of ln n for approximating set cover. *Journal of the ACM (JACM)* 45, 4 (1998), 634–652.

[2] Samir Khuller, Anna Moss, and Joseph Seffi Naor. 1999. The budgeted maximum coverage problem. *Information processing letters* 70, 1 (1999), 39–45.

[3] Jure Leskovec and Julian Mcauley. 2012. Learning to discover social circles in ego networks. *Advances in neural information processing systems* 25 (2012).

[4] Xin Sun, Xin Huang, and Di Jin. 2022. Fast algorithms for core maximization on large graphs. *Proceedings of the Very Large Data Bases (VLDB) Endowment* 15, 7 (2022), 1350–1362.

[5] Zhongxin Zhou, Fan Zhang, Xuemin Lin, Wenjie Zhang, and Chen Chen. 2019. k-Core Maximization: An Edge Addition Approach.. In *International Joint Conference on Artificial Intelligence (IJCAI)*. 4867–4873.

[6] Zhongxin Zhou, Wenchao Zhang, Fan Zhang, Deming Chu, and Binghao Li. 2022. VEK: a vertex-oriented approach for edge k-core problem. *World Wide Web* 25, 2 (2022), 723–740.